Improvement of Chat-like Dialogue Agent Satisfaction using Gamification

Arnaud JORDAN
Graduate School of Information Science and Technology
Hokkaido University
Sapporo, Japan
Email: arnaud@media.eng.hokudai.ac.jp

Kenji ARAKI
Graduate School of Information Science and Technology
Hokkaido University
Sapporo, Japan
Email: araki@ist.hokudai.ac.jp

***Abstract*—For many years, numerous dialogue agents have been developed. They have various objectives and performances. However, they generally focus only on dialogue handling or on accomplishing tasks. In this paper, we proposed a gamification of a previously proposed chat-like dialogue agent in order to make it more satisfying and enjoyable. Concretely, we provided some simple objectives to accomplish to the user in order to obtain game achievements.**

We carried out impression evaluation of the system before and after the gamification in order to check the influence of the proposed method. However, no satisfying impression improvement appeared. Nevertheless, we confirmed that users tend to complete provided objectives and to input more sentences in a system with gamification.

***Keywords*-dialogue processing, dialogue agent, gamification**

I. Introduction

Nowadays, dialogue agents are frequently used as personal assistant such as *Siri*[1], *Cortona*[2], *Amazone Alexa*[3] or to accomplish tasks such as *OK Google*[4]. In research field, ALICE [1], ELIZA [2] or the GA-ILS D system [3] are often cited. Nevertheless, in most of video games it is still not possible to communicate freely with characters we meet. Some commercial games make exception such as *Event[0]*[5] which uses a dialogue agent. However, this aspect of dialogue agent is not commonly addressed in current researches.

Although gamification has already been implemented in other fields with success notably in natural language processing [4]–[6], most of dialogue agents only provide conversation without any gaming concept. Nevertheless, there are some exceptions [7], but they not evaluate the influence of the gamification concepts. Moreover, we considered providing specific objectives can encourage the user to input sentences related to a specific topic which can be more easily predicted and as result covered by a corpus than completely free inputs.

That is why we attempted to make a dialogue agent more enjoyable and more satisfying using simple game achievements.

II. Outline

We used a previously proposed dialogue agent [8] as base. The flexibility of the system permits us to implement events representing the achievements of the game without adding specific processing. Concretely, we provided some additional dialogue samples containing *event tags* associated to sentences to the system as a learning corpus. Since the base system is able to provide several outputs to each input, it can output a response sentence and a tag for one single input.

Figure 1 shows an example of knowledge taught to the system.

U1: Go to sea
U2: OK, I go to sea#[a]<goToSea>

[a] # is used to separate outputs.

Figure 1. Example of knowledge tough to the system

Concretely, the system learns the dialogue sample and then if the user asks the robot to "*Go to sea*", the robot will answer "*OK, I go to sea*" and output the tag *<goToSea>*. When the robot outputs these kind of tags an achievement completion message is shown to the user (cf. Figure 6).

If the user input something related to "*Go to sea*" such as "*Please go to sea*" or "*Could you go to sea*", this will be assimilated to "*Go to sea*". However, if the input uses different words such as "*Let's go to the beach*", the system cannot match the sample. As a result, various dialogue samples are required to be efficient.

III. Setting of the system

Figure 2 shows the objectives that the users have to accomplish to obtain an achievement.

[1] https://www.apple.com/fr/ios/siri/
[2] https://www.microsoft.com/en/mobile/experiences/cortana/
[3] http://alexa.amazon.com
[4] http://ok-google.io/
[5] http://event0game.com/

1) Discovering the name of the robot
2) Discovering the robot age
3) Discovering the robot hobbies
4) Becoming friend with the robot
5) Making the robot get fat
6) Making the robot sing
7) Making the robot wear an astronaut helmet
8) Making it rain
9) Getting a camera from the robot
10) Making the robot go to the sea

Figure 2. Objectives to accomplish

Figure 3 shows the interface of the system. In order to make the dialogue agent more game-like, we added a graphic representation of the robot. In addition, some achievements change the robot appearance notably the seventh one.

Moreover, the interface uses responsive design[6] to suit smartphone screen as well as desktop one. In addition, we used *Web Speech API*[7] to provide voice recognition and voice synthesis.



Figure 3. System interface

The user is also able to send *emotions*[8] and *items*[9] (cf. Figure 4) to the robot. For example, if the robot receive a cake, it will say "*thank you*", but if it receive too many cakes, it will say "*I will get fat*".

[6]Using *Bootstrap* framework: https://getbootstrap.com.

[7]https://dvcs.w3.org/hg/speech-api/raw-file/tip/speechapi.html

[8]One of the six basic emotions proposed by Ekman [9]: joy, anger, disgust, sadness, surprise and fear.

[9]Camera and foods.

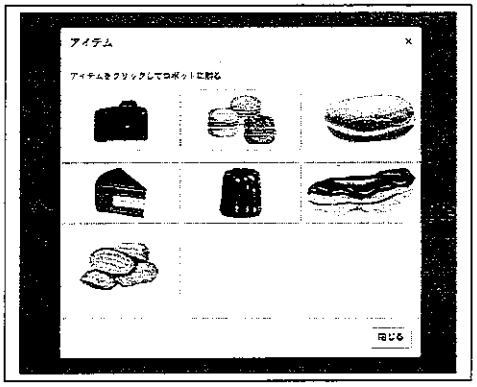


Figure 4. Items selection window

In addition, some objectives use the current state of the robot. For example, in order to achieve the objective: *getting a camera from the robot*, the user has to give a camera to the robot first. If he does not, the robot will answer "*I don't have camera*".

Figure 5 shows an example of knowledge tough to the system in order to use its current state. The system tends to select outputs which are in relation with the input and with its current state.

U1: Give me a camera
U2: OK, I give you a camera#<giveCamera>
<init>[a]
U1: Give me a camera
U2: Sorry, I don't have camera#<-giveCamera>[b]
<init>
U1: <camera>
U1: OK, I give you a camera#<giveCamera>
<init>
U1: <-camera>
U1: Sorry, I don't have camera#<-giveCamera>

[a]*<init>* tag indicates next dialogue has no relation with previous one.
[b]*<-giveCamera>* is the opposite tag of *<giveCamera>*.

Figure 5. Example of knowledge using state of the system

Concretely, when the system starts the *<-camera>* tag score is set to a defined value[10]. As a result, the system tends to select "*Sorry, I don't have camera*" as output since it is related to an active node. After the user gave a camera to the system, the *<camera>* tag score exceeds the *<-camera>* value and as consequence the system tends to select "*OK, I give you a camera*".

[10]Arbitrarily set to 5.

Figure 6 shows an example of achievement completion message displayed to the user after he accomplished an achievement.

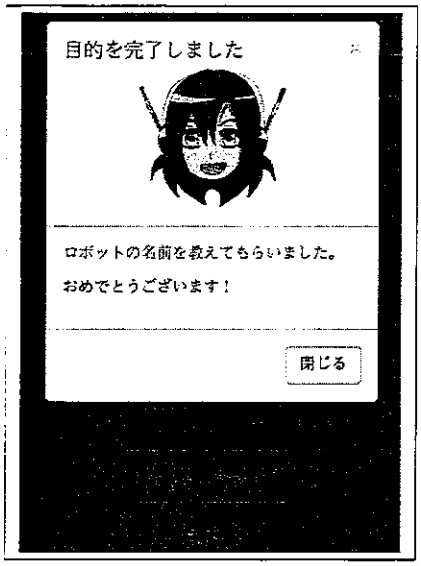


Figure 6. Achievement completion message

IV. Experiment

We carried out the experiment in order to evaluate the influence of gamification on the user impression. The experiment has been carried out in Japanese. The system without gamification is called *chat* and the one with gamification is called *game*. We used semantic differential approach [10], [11] with the same procedure as Uchida's paper [12]. We asked subjects to chat freely with the two systems, and we evaluated subject impression after they used each one. The first system is randomly chosen and we only provide some basic information about the system and how to use it.

We ask subjects to chat with the system until they lose interest. Subjects used a smartphone's keyboard to chat with the system. After using one system, the subject fills out the corresponding survey form.

We also ask three questions as below:

- Do you know dialogue agent?
- Do you already use a dialogue agent? If you did, which one?
- Do you prefer typing input sentence or use voice recognition?

A. Subjects

Table I shows information about participants of the experiment. 6 subjects participated to the experiment, all of them were Japanese native speakers.

Table I
Subject's information

	Number
Subject	6
Male	3
Female	3
Student	5
Worker	1
Average age	21.17

B. Results

As shown on Table II, most of the subjects already used a dialogue agent even if they do not really know what exactly it is. The most frequently cited systems were *Siri* and *Pepper*[11]. It is possible to compare this result with our previous paper [13]. We can notice that even if dialogue agent are not well known yet, they are more often used.

Table II
Experiment's result

Question	Yes (subject)
Know dialogue agent	2
Used dialogue agent	5
Prefer voice input	3

Figure 7 shows an example of dialogue with the game system.

U[a]: 名前は何ですか
(What is your name?)
S[b]: 名前はセシールです
(My name is Cecile)
U: 趣味は何ですか
(What are your hobbies?)
S: 趣味は水泳です
(My hobby is swimming)
U: カメラが欲しいなあ
(I want a camera)
S: カメラを持ってないですよ
(I do not have camera!)
S: <-giveCamera>[c]
U: 友達になりましょう
(Let's be friend)
S: 友達になろう
(Let's be friend)
S: T:<friend>

[a]User input
[b]System output
[c]The robot is not able to give camera

Figure 7. Example of dialogue with the game system

[11]https://www.softbank.jp/en/robot/

We can see that the user focused on the objectives and did not attempt to have a natural dialogue with the system. The conversation subject change from the *swimming* to *camera* suddenly without any reason.

Table III shows the mean number of sentences input by the users during the experiment.

Table III
NUMBER OF INPUTS

	Chat	Game
Number of inputs	47.2	52.5

Table IV shows the completion of each objective of the game. On average each subject completed 6.5 objectives.

Table IV
OBJECTIVES COMPLETION

Objective	Completion (subject)
1	3
2	4
3	4
4	5
5	6
6	6
7	2
8	3
9	4
10	2
Average	3.9
Total	39

Table V and Figure 8 show the result of the impression evaluation. The result are rated from 0 (negative) to 6 (positive).

Table V
EXPERIMENT'S RESULT

Adjectives		Chat	Game
Bad	Good	4.00	3.33
Odious	Cute	4.17	4.50
Boring	Interesting	3.17	3.83
Malicious	Kind	3.33	2.67
Idiot	Clever	3.33	2.67
Inconvenient	Useful	3.00	2.83
Won't use	Want to use	3.50	3.83
Empty	Full	2.83	3.50
Tedium	Have many interest	4.67	4.50
Complicated	Simple	3.17	3.00
Machine-like	Human-like	1.67	1.67
Slow	Fast	3.50	3.00
Hard to be intimate	Easy to be intimate	2.50	3.33
Hard to understand	Easy to understand	3.83	3.33
Disappointed	Satisfied	4.00	2.66

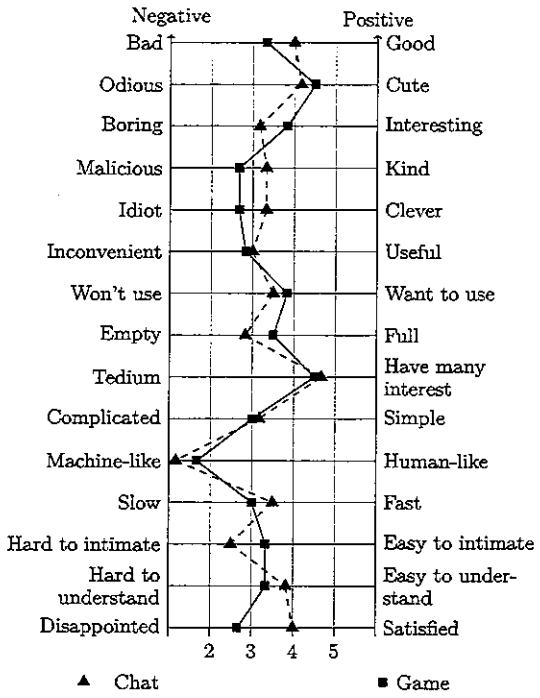


Figure 8. Semantic differential experiment's results

C. Consideration

The system without gamification obtained a better mean result 3.38 than the system with gamification 3.24. Nevertheless, the game system was a little more interesting 3.83 than the chat system 3.17. However, the majority of the users are more satisfied by the system without gamification 4.00 against 2.66. We consider this is because the game objectives were not enough attracting and some achievements were hard to accomplish making user disappointed. Therefore, we think the game proposed to the user must be challenging and the reward must be attracting to influence positively the user.

In addition, even if the impression was worse, subjects input more sentences in the system with gamification, 52.5 average inputs against 47.2 for the system without gamification. Moreover, 65% of objectives were completed. Therefore, we think gamification can be a good solution to increase the length of dialogue between human and machine.

Furthermore, we consider difference between the two systems are not significant. However, a simple gamification does not provide a significant advantage to a dialogue agent.

In another hand, some users feel hard to use the

system on small screen of smartphone since it is not possible to see the robot, the dialogue and the keyboard at the same time. In order to solve this problem, we consider to carry out experiment on laptop or to use vocal input.

V. Conclusion

In conclusion, the gamification can be applied to dialogue agents, however as any other game the objectives must be enough motivating for the user, otherwise, it can have a negative impact. In particular, we consider the awards provided to the users when they accomplish an objective must attracting in order to satisfy them.

Moreover, users naturally attempt to complete provided objectives. Therefore, since users focus on objectives, objectives orient the content of the user input and this can make dialogue less natural. Concretely, users tend to not reply to the system outputs and only focus on complete objective list. That is why we consider adding new concept in the game to encourage user to reply to system outputs. For example, replying to the system outputs can increase the score of the user.

In addition, since the base system is *language-independent*, the proposed system can be simply adapted to other language without changing the system itself but only its interface and the dialogue samples.

In further research, we will attempt to enhance the game in order to make it funnier and attracting for the user. For example, we think we can add score and special missions to accomplish with the robot such as defeat an enemy. Furthermore, we also consider implementing experience points in order to encourage user to chat more longer with the system. Similarly to role-playing video game, more experience the robot has, more it will be able to accomplish specific actions.

Moreover, adding interactions with other users can also be a good solution to increase the interest of the user. For example, the state of the robot can be shared between all the users, if one of them give a camera, then the robot can give it to another one. As a result, some achievement could require the cooperation between several users.

References

[1] R. S. Wallace, "The anatomy of A.L.I.C.E.," *Parsing the Turing Test*, vol. pp. 181-210, 2009.

[2] J. Weizenbaum, "ELIZA - a computer program for the study of natural language communication between man and machine," *Communications of the ACM 9 (1)*, vol. 36"45, 1966.

[3] K. Araki and M. Kuroda, "Generality of spoken dialogue system using sega-il for different languages.," in *Computational Intelligence*, pp. 72–77, 2006.

[4] N. Venhuizen, V. Basile, K. Evang, and J. Bos, "Gamification for word sense labeling," in *Proc. 10th International Conference on Computational Semantics (IWCS-2013)*, pp. 397–403, 2013.

[5] J. Bos and M. Nissim, "Uncovering noun-noun compound relations by gamification," in *Proceedings of the 20th Nordic Conference of Computational Linguistics, NODALIDA 2015, May 11-13, 2015, Vilnius, Lithuania*, no. 109, pp. 251–255, Linköping University Electronic Press, 2015.

[6] C.-C. Musat Thisone, A. Ghasemi, and B. Faltings, "Sentiment analysis using a novel human computation game," in *Proceedings of the 3rd Workshop on the People's Web Meets NLP: Collaboratively Constructed Semantic Resources and their Applications to NLP*, pp. 1–9, Association for Computational Linguistics, 2012.

[7] L. Benotti, M. C. Martínez, and F. Schapachnik, "Engaging high school students using chatbots," in *Proceedings of the 2014 conference on Innovation & technology in computer science education*, pp. 63–68, ACM, 2014.

[8] A. JORDAN and K. ARAKI, "Real-time language-independent algorithm for dialogue agents," 知能と情報, vol. 28, no. 1, pp. 535–555, 2016.

[9] P. Ekman, "Universals and cultural differences in facial expression of emotion," *J. Cole ed. Nebraska Symposium on Motivation.*, pp. 207–283, 1972.

[10] D. R. HEISE, "The semantic differential and attitude research," *Attitude Measurement*, pp. 235–253, 1970.

[11] K. Anzou, K. Murata, and M. Numasaki, "Shakai Shinrigaku Kenkyuu Nyuumon," September 2009.

[12] Y. Uchida and K. Araki, "Evaluation of a System Noun Concepts Acquisition from Utterances about Images (SINCA) Using Daily Conversation Data," *Proceedings of NAACL-HLT*, pp. 65–68, 2009.

[13] A. Jordan and K. Araki, "Spoken dialog processing for acquiring taste and knowledge," *Proceedings of PACLING2013*, 2013.